Geophys. J. Int. (2000) **140,** 598–610

New palaeomagnetic result from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vendian

S. A. Pisarevsky,[1] R. A. Komissarova[2] and A. N. Khramov[2]

[1] *Tectonics Special Research Centre, Department of Geology and Geophysics, the University of Western Australia, Nedlands,* WA 6907, *Australia. E-mail: sergei@tsrc.uwa.edu.au*

[2] *All Russian Petroleum Research and Geological Exploration Institute (VNIGRI), Liteiny* 39, 191104 *St. Petersburg, Russia*

Accepted 1999 October 14. Received 1999 October 1; in original form 1999 May 31

SUMMARY

A palaeomagnetic study of Vendian red sediments from the Lena River section on the western margin of Lake Baikal in the region of Cisbaikalia (54°N, 108°E) has isolated a stable remanence with direction $D = 296.3°$, $I = -27.7°$ (high-temperature component) and a corresponding pole of 2.7°S, 168.2°E. The primary nature of this remanence is confirmed from a positive fold test, dual polarities and the presence of detrital haematite. This result, together with all late Precambrian–Early Cambrian palaeomagnetic data from Siberia, indicates that Siberia occupied low latitudes during that time. It has been proposed on the basis of palaeomagnetic data that Laurentia occupied high latitudes during the Vendian, so it would appear that there cannot have been any Laurentia–Siberia connection at that time. A review of Vendian to Cambrian Laurentian palaeomagnetic data shows that such an interpretation is ambiguous. An alternative interpretation places Laurentia in low latitudes and confirms the Laurentia–Siberia fit of Hoffman (1991) and Pelechaty (1996). However, the lack of Late Vendian palaeomagnetic data for Siberia still allows the possibility that it could have occupied high latitudes during that time.

Key words: palaeomagnetism, sediments, Siberia, Vendian.

1 INTRODUCTION

During recent years various authors have postulated the existence of a Neoproterozoic supercontinent—Rodinia (e.g. McMenamin & McMenamin 1990; Dalziel 1991; Hoffman 1991; Powell *et al.* 1993). Although there is general agreement on the existence of such a supercontinent, the details about the reconstruction of its constituent parts and its palaeolatitudinal history are still subjects for discussion. In particular, the position of Siberia in Rodinia is the subject of some debate. Probably the most common point of view is the connection of Siberia to northern Laurentia (Hoffman 1991; Condie & Rosen 1994; Pelechaty 1996; Frost *et al.* 1998; Rainbird *et al.* 1998). However, the details of this fit are quite different in these reconstructions. More detailed late Precambrian and Cambrian palaeomagnetic data from Siberia would provide a means of distinguishing between these reconstructions, as they would provide palaeolatitudinal and azimuthal positions for comparison with other continents.

There are still only a few palaeomagnetic data from Vendian rocks of Siberia (Smethurst *et al.* 1998). Only four results are available, two with an assumed age of 590 Ma and two with an age of 650 Ma. These results have reliability criteria (Van der Voo 1988) of 3 or 4. The common feature of all available Precambrian and Cambrian palaeomagnetic data from Siberia is their relatively low palaeolatitude. This agrees with the assumption of a Laurentia–Siberia connection for the Late Riphean and Cambrian, because the available palaeomagnetic data for these epochs from Laurentia also support a low-latitude position (Torsvik *et al.* 1996). However, Torsvik *et al.* (1996), on the basis of selected palaeomagnetic data, proposed that Laurentia was at high latitudes during the Vendian. Therefore, the main aim of this study is to provide new palaeomagnetic results from Vendian rocks of Siberia to indicate whether the Laurentia–Siberia connection persisted through this time.

2 GEOLOGY AND SAMPLING

The studied area (Fig. 1) is a part of Cisbaikalia, the region to the west of Lake Baikal and near the southern boundary of the Siberian craton. The early Precambrian metamorphic basement is covered here by the Riphean Baikalian succession made up of the Goloustnaya, Uluntui and Kachergat Formations from

© 2000 RAS

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

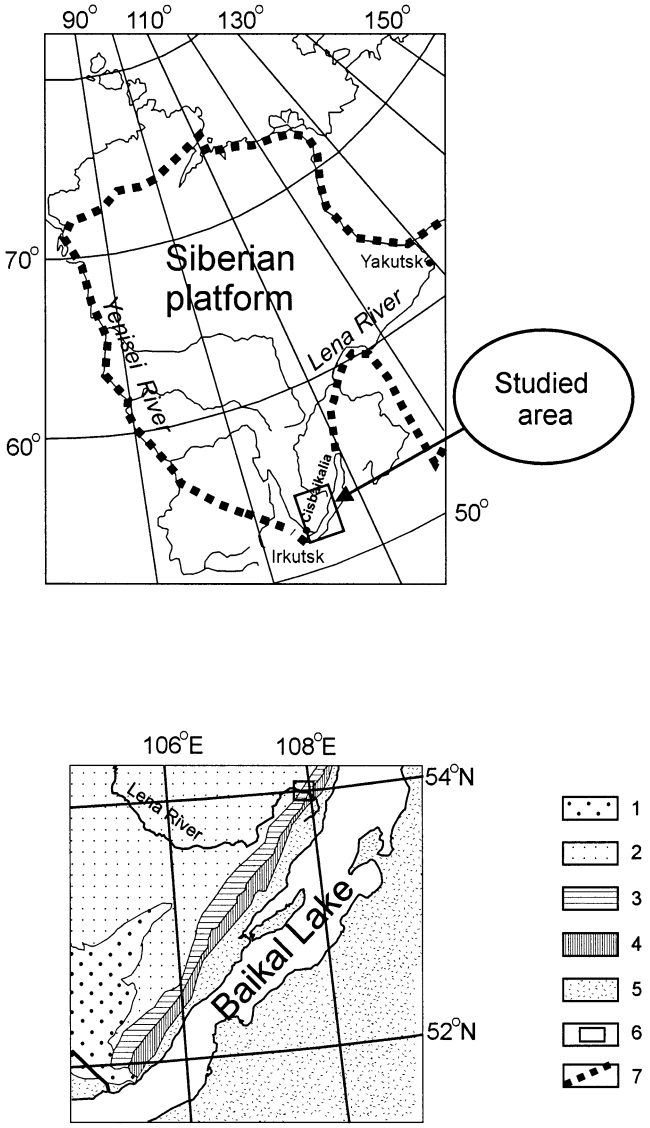


Figure 1. Sketch map of Siberia and simplified geological map of Cisbaikalia. 1: Mesozoic rocks; 2: Early Palaeozoic rocks; 3: Vendian (Yudomian) complex; 4: Baikalian (Riphean) complex; 5: Early Precambrian metamorphics; 6: sampling area; 7: platform boundary.

the base to the top. These sediments were strongly deformed, probably during the early stage of the Baikalian tectonic epoch at about 850 Ma (Khain 1994). The Vendian succession in the study area includes, in its upper part, a 200 m thick sequence of red clastic sediments that were sampled for palaeomagnetic study. According to Trunov (1975) and Salop (1973) these red beds correlate with the Moty Formation, the youngest formation of the Yudomian (Vendian) sequence of Siberia. However, Khomentovsky (1985) suggested that the red beds belong to the upper part of the Ushakovka Formation, which overlies the Kachergat Formation with local unconformity. Radiometric ages of unknown method and analytical precision are cited by Khain & Bozhko (1988) as 644 Ma for the Ushakovka Formation and 606–609 Ma for the Moty Formation. The red bed sequence, sometimes referred to as molasse (Khain 1994), is overlain by Lower Cambrian dolostones with a prominent angular unconformity. Although the isotopic dates for the Vendian succession may in themselves be only approximations to the true values, they are in general agreement with the regional stratigraphic relationships, which suggest a Vendian age somewhere between 650 and 550 Ma. The K–Ar date of 585 Ma (glauconite) at the top of the Ayankan suite (Khomentovsky 1985), which corresponds to the upper part of the Moty Formation, obtained from Ushakovka river 300 km southwest of the studied area permits us to reduce this time interval correspondingly.

The red bed sequence strikes northeasterly between near-parallel folds ranging from metres to hundreds of metres in size. The angles of dip vary from 20° to ~90°, but in most cases are around 45°. This folding is probably caused by the collision of the Olkhon terrane and the Siberian craton between 500 and 450 Ma (Fedorovsky 1997). The red sequence was sampled in two parallel sections along the Greater Lena River and along Sokhatiny creek. 80 oriented hand samples from 10 outcrops were collected. A magnetic compass was used for orientation. 2–6 cubic specimens with sides of 2 cm were trimmed from each oriented sample.

3 LABORATORY METHODS AND MAGNETIC PROPERTIES

The palaeomagnetic study of the collection has been carried out mainly in the palaeomagnetic laboratory of the Tectonics Special Research Centre at the University of Western Australia. Some measurements were also made in the palaeomagnetic laboratory of VNIGRI (St. Petersburg, Russia). Magnetic remanence was measured using a 2G755R cryogenic magnetometer and JR4 spinner magnetometer. The majority of specimens were subjected to stepwise thermal demagnetization using MMTD2 furnaces manufactured by Magnetic Measurements (~10 nT residual field) and a VNIGRI home-made screened furnace (~30 nT residual field). A few specimens were subjected to stepwise alternating field (AF) demagnetization using a 2G600 automated degaussing system, but this type of magnetic cleaning proved to be ineffective for the rocks studied. Magnetic susceptibility was measured after each heating step with an MS2 susceptibility meter or a KLY-2 bridge. Isothermal remanent magnetization (IRM) experiments were carried out using an MMPM9 pulse magnetizer. The natural remanent magnetization (NRM) of the red sediments studied ranges from 0.7 to 4.1 mA m^{-1}, and their magnetic susceptibility from 0.5 to 4.0×10^{-4} SI units.

Some typical examples of changes in magnetic susceptibility during thermal demagnetization are shown in Fig. 2. In most cases the susceptibility increases strongly at 450 °C or higher, although in some cases this increase may commence at temperatures as low as 300 °C. Usually this increase slows down or stops at temperatures higher than 600 °C. There are different ways to explain this effect, which is often observed in red sediments (e.g. Dunlop 1972; Lovlie *et al.* 1984; Dunlop & Ozdemir 1997). For example, it can be caused by the transformation of goethite into haematite or by the generation of magnetite due to the breakdown of some non-magnetic mineral (Dunlop & Ozdemir 1997, pp. 446 and 456). It is also possible that different processes are working at the same time. Our attempts to resolve this problem are discussed below. However, the most important fact for the purposes of this study is that such mineralogical changes sometimes introduce significant noise into the palaeomagnetic analysis. In spite of the small residual

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

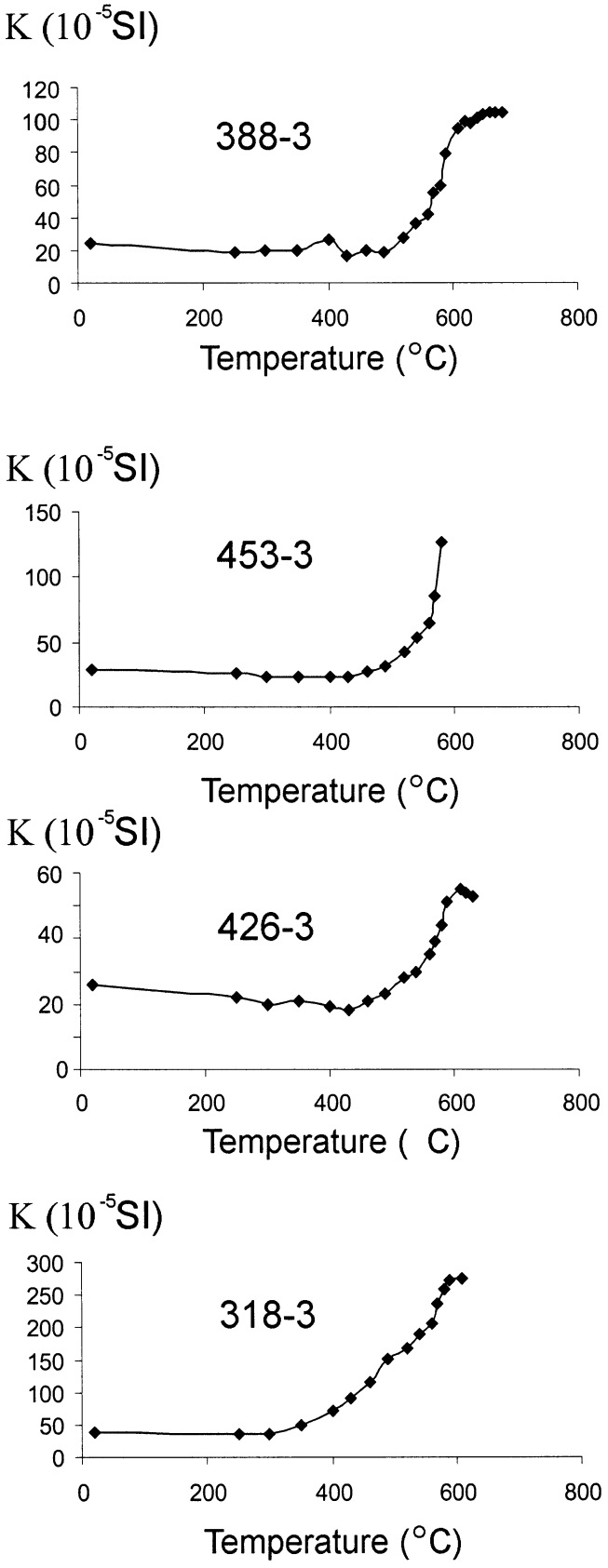


Figure 2. Examples of increasing magnetic susceptibility during thermal demagnetization.

fields, irregular parasitic magnetizations were observed in some specimens during thermal treatment at high temperatures.

Three examples of IRM saturation curves before and after heating to 690 °C are shown in Fig. 3. Before heating (Fig. 3a) the curves show evidence in favour of the dominance of high-coercivity magnetic minerals such as haematite or goethite. The sharp increase in the IRM in the range 200–600 mT may be caused by the presence of detrital haematite. Because saturation is not achieved in fields of even ~3 T, authigenic haematite and/or goethite may also be present (Dunlop & Ozdemir 1997).

The same three samples were used for the analysis of rock magnetic mineralogy following the method proposed by Lowrie (1990). Fields of 2900, 400 and 120 mT were applied successively along three orthogonal axes of each sample. These specimens were then subjected to stepwise thermal demagnetization. This procedure enables the detection of three groups of minerals: (1) authigenic haematite and goethite, (2) pyrrhotite, single-domain magnetite and probably detrital haematite (Dunlop & Ozdemir 1997, p. 445), and (3) low-coercivity magnetite. The results are shown in Fig. 4, and in each case there is no significant feature at 120 °C. This allows us to reject goethite as a possible accessory mineral. There is also no evidence for the presence of pyrrhotite or maghemite. Curves (b) and (c) show the presence of some minor amounts of magnetite, but it is obvious that haematite is the dominant magnetic mineral in these sediments. An important observation is the high percentage of a moderate-coercivity mineral with a Curie temperature of ~680 °C, which suggests the presence of detrital haematite.

The IRM curves for these three specimens were repeated after thermal demagnetization (Fig. 3b). Heating causes a significant change in shape of the IRM curves. At low fields there is a rapid increase in IRM, which suggests the generation of some low-coercivity mineral (probably magnetite). However, the value of IRM at the maximum field of ~3 T remains virtually unchanged from that of the samples prior to heating. Therefore, this low-coercivity mineral is not a significant part of the total IRM. This contrasts with the results of Torsvik *et al.* (1985) from Devonian red sediments of Spitsbergen. Another obvious difference after heating is that there has been a redistribution of the coercivities of the haematite particles to higher values. Dunlop (1972) has reported a similar result for some red sediments from France. These observations provide some explanation for the dramatic rise in magnetic susceptibility during thermal demagnetization (Fig. 2). The main conclusion from these rock magnetic experiments is that there is some evidence for the presence of detrital haematite in these red sediments.

Measurement of the anisotropy of magnetic susceptibility (AMS) was carried out on 16 specimens. The results are shown in Figs 5(a) (before tilt correction) and (b) (after tilt correction). A better grouping of the minimum axes near the palaeovertical is observed after tilt correction. The maximum axes are scattered near the palaeohorizontal plane. Such a distribution is characteristic of a depositional or early diagenetic magnetic fabric (Tarling & Hrouda 1993).

4 PALAEOMAGNETIC ANALYSIS

The characteristic components of NRM (ChRM) were isolated using a least-squares algorithm (Torsvik 1986), combined with the analysis of stereograms. Fisher (1953) statistics were

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

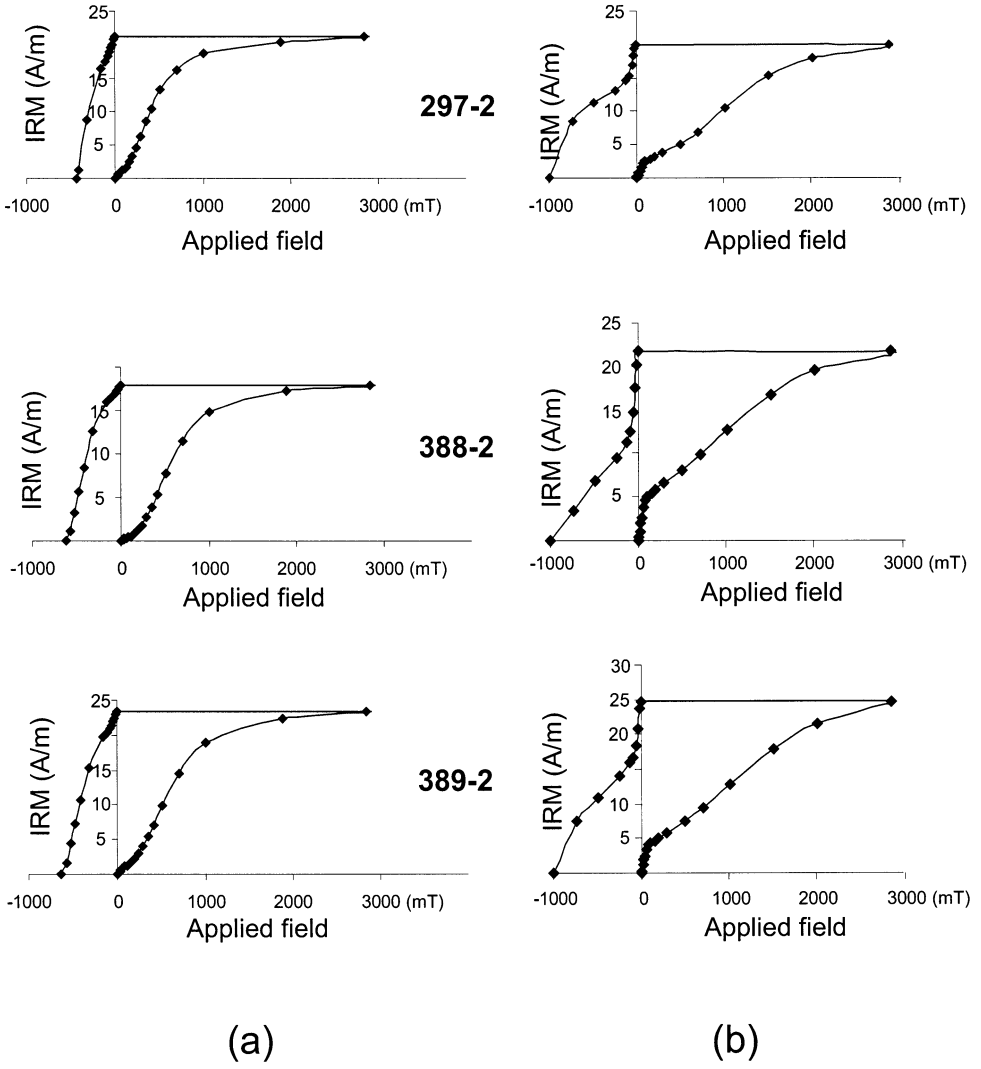


Figure 3. IRM saturation curves (a) before thermal demagnetization and (b) after thermal demagnetization; this process is shown in Fig. 4.

used for calculations. Some typical examples of the thermal demagnetization are shown in Fig. 6. The first temperature step (100 °C) usually destroyed the 'laboratory' viscous component. The present-day field (PDF) component was isolated at temperatures below 210 °C. Fig. 7 shows the distribution of this component (*in situ*). In some cases (Figs 6d and e) it comprises more than 80 per cent of the NRM. One or two components were isolated during further demagnetization.

The high-temperature component (usually with unblocking temperatures higher than 420°) has dual polarity, although there are only three samples with normal polarity in the lower part of the section (for Palaeozoic palaeomagnetic data from Siberia the normal-polarity direction is SE with negative inclination). However, the reversal test (McFadden & McElhinny 1990) is indeterminate due to the large scatter. The stereoplots are shown in Figs 8(a) (before tilt correction) and (b) (after tilt correction). Table 1 shows the corresponding sample means and statistics. In this case the directions are better grouped after tilt correction. Three variants of the fold test were performed. The fold test of McFadden (1990) is positive at the 99 per cent confidence level. The cluster fold test of McFadden (1998), which is probably the most rigorous one, shows the maximum grouping of the high-temperature component at 100 per cent unfolding, with the probability of exceeding the observed value of $f = 2.43$ being 3.74 per cent. The corresponding values for the case of 0 per cent unfolding are $f = 19.01$ and 0.00 per cent. This result cannot be interpreted as a positive fold test, but shows some possibility of the prefolding nature of the high-temperature component. The fold test of Tauxe & Watson (1994) gave maximum eigenvalues between 76 and 102 per cent of unfolding (Fig. 9). Together, these results allow us to assume the prefolding, and hence pre-Early Cambrian, nature of the high-temperature component.

In the medium-temperature interval (usually below 450°) many Zijderveldt plots show some evidence of the possible presence of an additional remanence component (Figs 6e, a and b). This 'medium-temperature component' may be artificial due to the overlapping of the blocking temperature spectra of the low- and high-temperature components. Another possibility is the coincidence of a high-temperature Vendian component and NNW–SSE bipolar Cambrian magnetization. However, there were specimens where it was impossible to isolate the high-temperature component due to the random behaviour of magnetization caused by the generation of a new magnetic

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

Table 1. Sample mean palaeomagnetic directions from Vendian red sediments of Lena River Section, Cisbaikalia, Siberia, Russia (54.0°N, 108.0°E).

Component of magnetization	N/n (°)	Decl. (°)	Incl. (°)	k (°N)	α_{95} (°E)	Plat (°)	Plong (°)	D_p	D_m
1 High-T, tilt corr.	32/39	296.3	−27.7	8.6	9.2	−2.7	168.2	5.5	10.1
2 High-T, *in situ*	32/39	285.9	−9.0	3.4	16.4	−5.5	182.4	8.4	16.6

N/n = number of samples/specimens; Decl, Incl = sample mean declination, inclination; k = best estimate of the precision parameter of Fisher (1953); α_{95} = the semi-angle of the 95 per cent cone of confidence; Plat, Plong = latitude, longitude of the palaeopole; D_p, D_m = the semi-axes of the cone of confidence about the pole at the 95 per cent probability level.

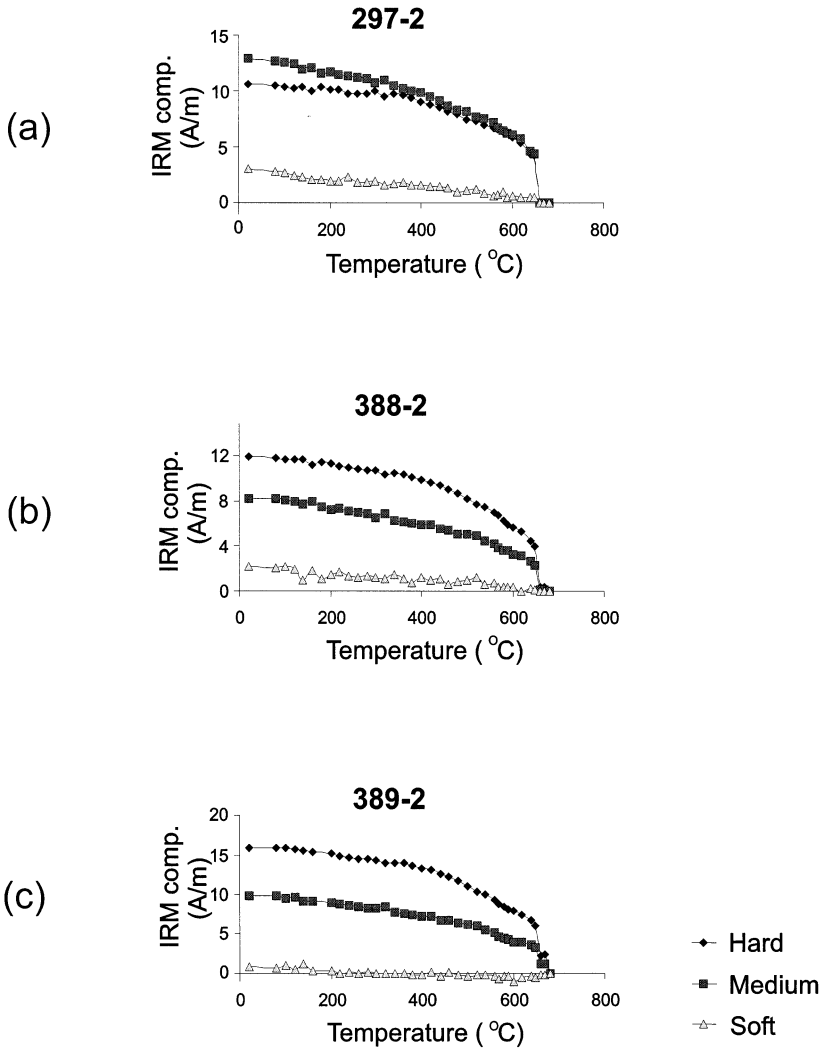


Figure 4. Three-axis IRS(T) curves (Lowrie 1990). The magnetization fields are 2.9 T (hard), 0.4 T (medium) and 0.12 T (soft).

mineral (see Section 3), so we have many cases when only this direction was retrieved. Thus, we cannot completely reject the possibility of the geomagnetic nature of this component. Fig. 10(a) shows a stereoplot of the medium-temperature component *in situ*, and Fig. 10(b) the same component after tilt correction. Both distributions are obviously non-Fisherian. The fold test of Tauxe & Watson (1994) gives scattered eigenvalues with a flat maximum between 9 and 69 per cent of unfolding. The cluster test of McFadden (1998) gives an inconclusive result with the maximum grouping at 35 per cent of unfolding ($f = 3.84$; the probability of exceeding this value is 0.19 per cent). The corresponding distribution is shown in Fig. 10(c). All these arguments favour the artificial, superpositional nature of the 'medium-temperature component', and we do not use it in further discussion.

5 DISCUSSION

The primary nature of the high-temperature component is supported by several factors. These are the positive fold test, dual polarity, rock magnetic evidence for the presence of detrital haematite and observations of the magnetic fabric. The

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

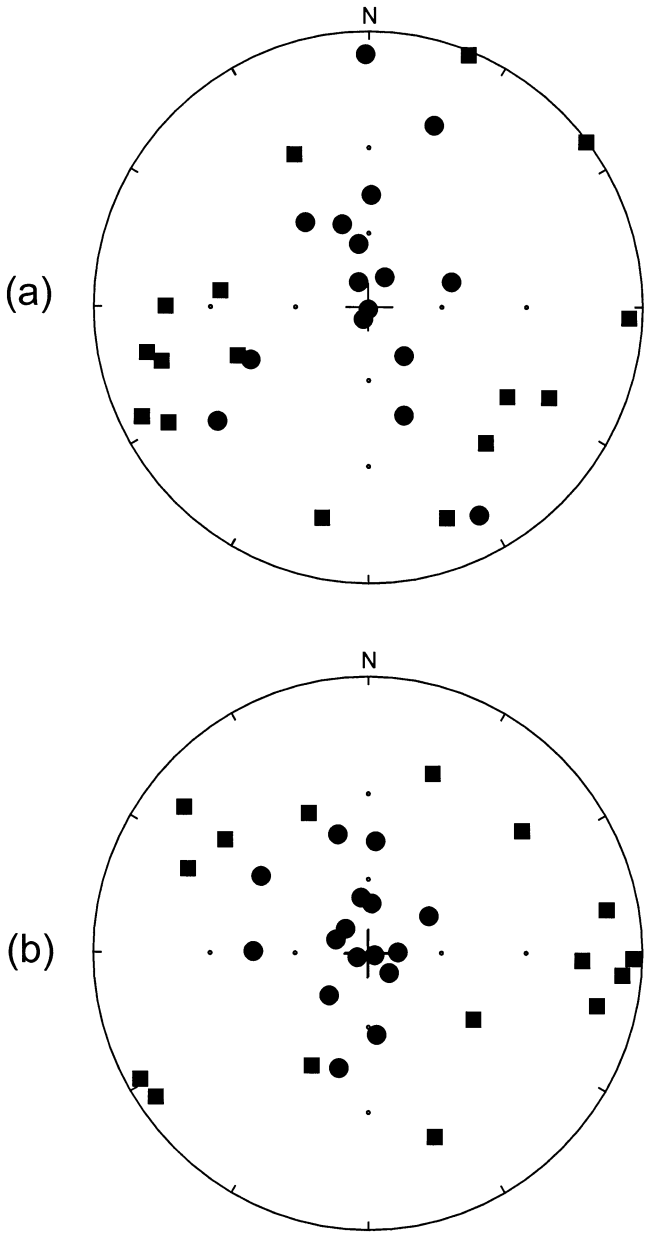


Figure 5. Stereoplots (equal-angle projection) of maximum (squares) and minimum (circles) AMS axes: (a) before tilt correction; (b) after tilt correction.

relatively large scatter of the directions may be partly caused by the spectrum of grain sizes of the magnetic particles (wide range of coercivities and blocking temperatures; see Figs 3, 4 and 5) and the dynamic environment during the deposition of the molasse sediments. Lovlie *et al.* (1984) and Torsvik *et al.* (1985) found a similar effect during their palaeomagnetic study of Devonian red beds in Spitsbergen. They suggested that the hydrodynamic conditions during deposition might have affected the alignment of the haematite particles due to their wide range of grain sizes. The presence of authigenic haematite, which probably carries secondary magnetization (see previous section), may also be a cause of the increased scatter. This is because the blocking temperature spectra of high- and what we call 'medium-temperature' (see Section 4) components partly overlap, and the final high-temperature component may still be contaminated. We believe that any such contamination is relatively minor, and regardless of the actual magnetic carrier of the high-temperature component or its mode of acquisition, the positive fold test indicates a Vendian remanence.

The north palaeopole calculated from the direction of the high-temperature component (2.7°S, 168.2°E) lies at ∼18° distance from the two 650 Ma poles of the Ust-Kirbinsk red sandstones and ∼7° from the 730 Ma pole of the Kandyk sediments and intrusives (Pavlov 1994) (Table 2). Such proximities can be explained either by very slow plate movement during Late Riphean and Vendian times, or by a remagnetization of Kandyk rocks. There is no direct evidence of such remagnetization, but Smethurst *et al.* (1998) mentioned that these rocks were cut by intrusions of the Ingili massif dated at between 640 and 690 (K–Ar on biotite) and 650 Ma (Pb–Pb). Recently, Rainbird *et al.* (1998) established a much older age for the Kandyk Formation (∼1000 Ma), so, from our point of view, the probability of Vendian remagnetization of these rocks increases. Whatever the correct interpretation, all these palaeomagnetic data show that Siberia occupied low latitudes between 650 and 580 Ma. Early Cambrian palaeomagnetic data (Pisarevsky *et al.* 1997; Smethurst *et al.* 1998) also restrict the palaeoposition of Siberia to low or at most moderate latitudes, and with the exception of the enigmatic result of Kirschvink & Rozanov (1984), in a similar azimuthal orientation (Table 2). This palaeomagnetically determined near-equatorial position is entirely consistent with the abundant carbonate sedimentation across the Siberian platform in Yudomian (Vendian) and Cambrian times.

Table 2. Selected Early Cambrian, Vendian and Late Riphean palaeomagnetic poles from Siberia.

N	Rock unit	Age Ma	Location		North pole		Reference
			°N	°E	°N	°E	
1	Inican sediments	538	59	135	−46	162	Osipova in Smethurst *et al.* (1998)
2	Pestrotsvetna Group	538	60	135	−46	181	Osipova in Smethurst *et al.* (1998)
3	Kessyusa Formation	545	71	123	−38	165	Pisarevsky *et al.* (1997)
4	Lena red sediments	580–650	54	108	−3	168	This study
5	Ust-Kirbinsk red siltstone	650?	58	137	−19	175	Pavlov in Smethurst *et al.* (1998)
6	Ust-Kirbinsk red siltstone	650?	59	137	−20	180	Pavlov in Smethurst *et al.* (1998)
7	Kandyk sediments and intrusives	730*	59	136	−4	175	Pavlov in Smethurst *et al.* (1998)

* The age of the Kandyk Formation was recently revised by Rainbird *et al.* (1998).

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

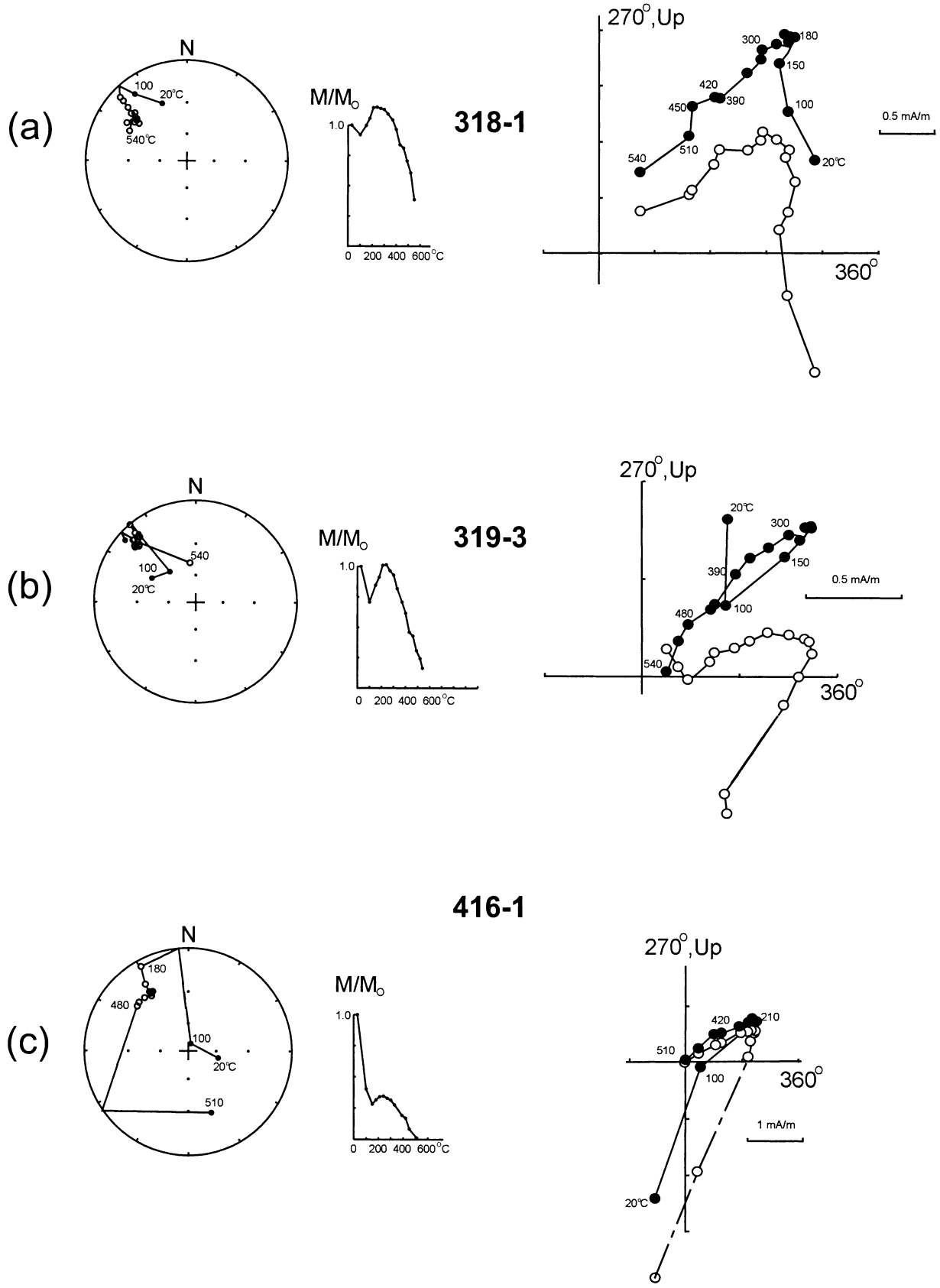


Figure 6. Examples of thermal demagnetization [equal-angle projections, normalized demagnetization curves and orthogonal plots (Zijderveld 1967)]. All results are in the stratigraphic system (after tilt correction). Solid and open symbols on the stereoprojection represent downward and upward directions, respectively; solid and open symbols on the Zijderveld diagram denote horizontal and vertical projections, respectively.

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

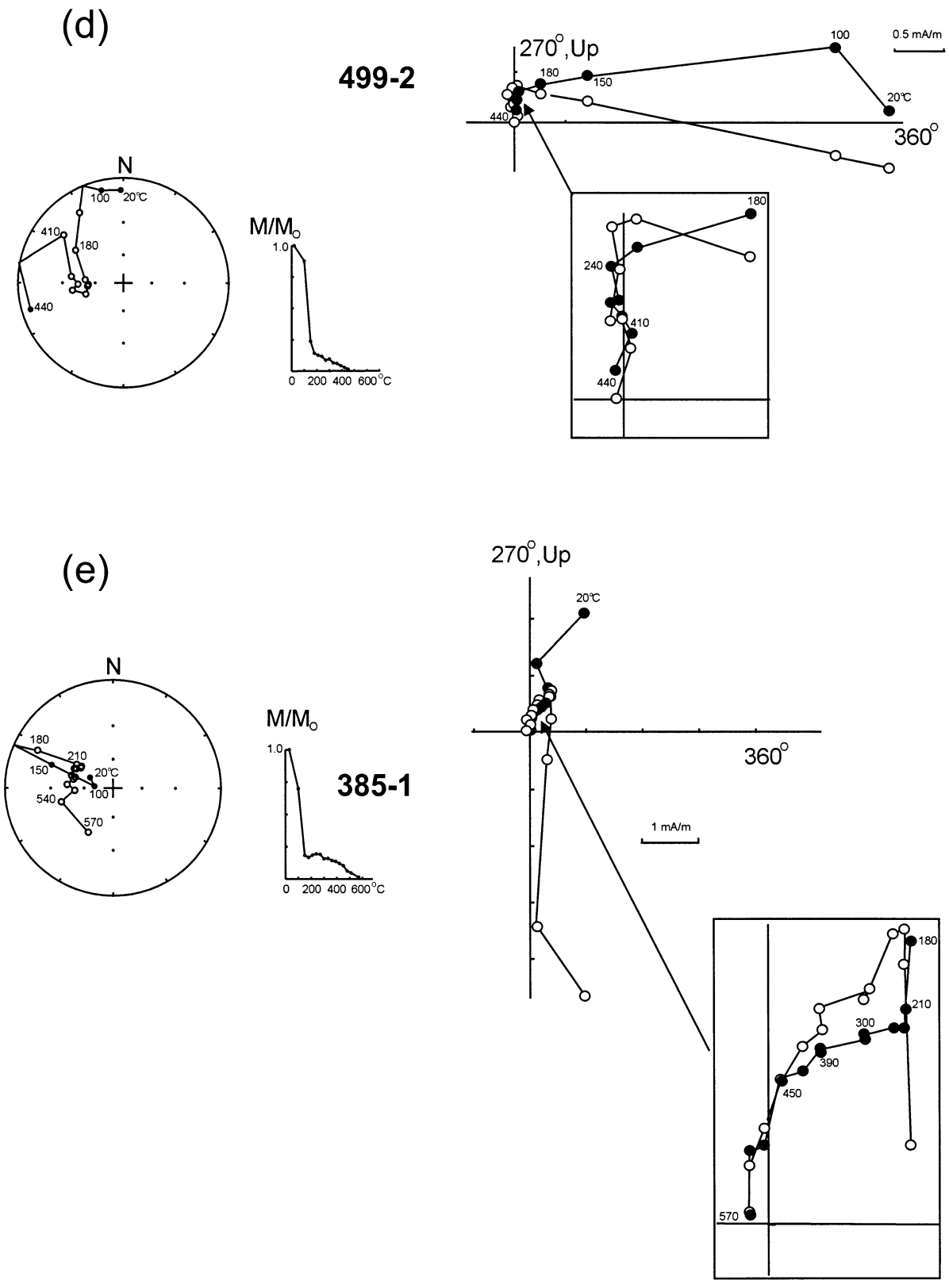


Figure 6. (*Continued.*)

There are several lines of evidence for the connection between Siberia and the northern margin of Laurentia in the Late Riphean and Vendian and probably the Early Cambrian (Hoffman 1991; Condie & Rosen 1994; Pelechaty 1996; Frost *et al.* 1998). Palaeomagnetic data from Siberia show that it was in low latitudes throughout this time interval. Palaeomagnetic results also support a low-latitude position ($<35^\circ$) for Laurentia during the Late Riphean ($\sim$750 to $\sim$650 Ma) and during the

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

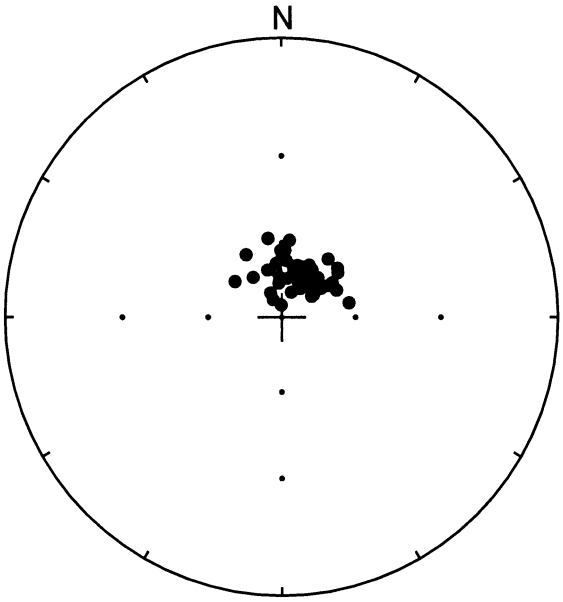


Figure 7. Stereoplot (equal-angle projection) of the low-temperature (PDF) component in present coordinates. See caption to Fig. 6.

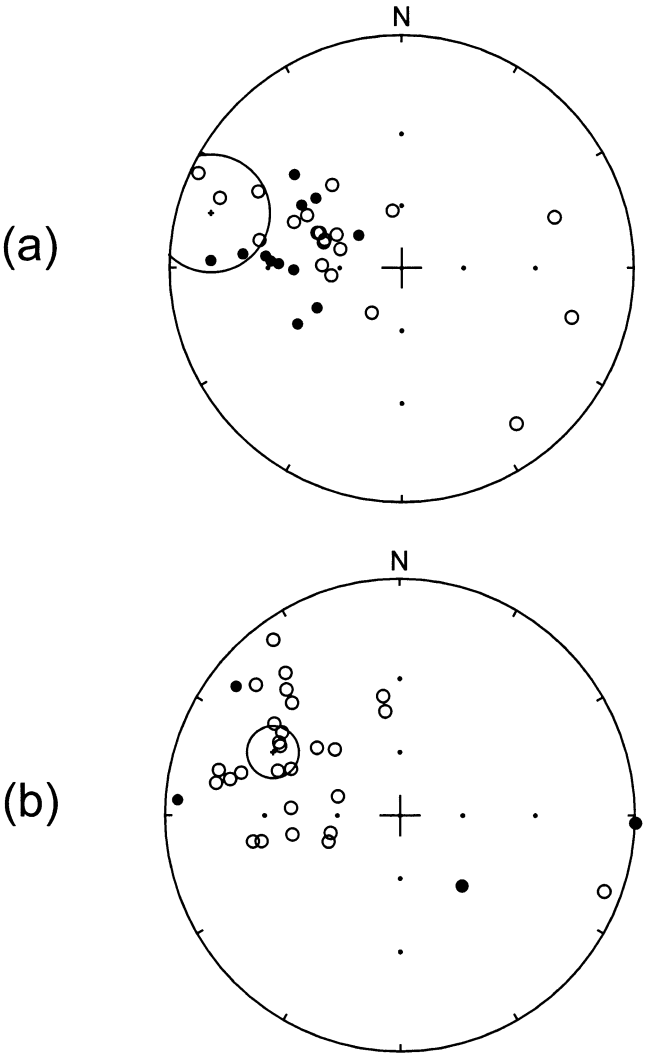


Figure 8. Stereoplot (equal-angle projection) of the high-temperature component: (a) *in situ*; (b) after tilt correction. See caption to Fig. 6.

latest Vendian to Cambrian (Dalziel 1997; Torsvik *et al.* 1996). However, several authors (e.g. Symons & Chaisson 1991; Powell *et al.* 1993; Park 1994; Torsvik *et al.* 1996; Kirschvink *et al.* 1997) have suggested a high-latitude position for Laurentia at ~580 Ma. This conclusion is based on four palaeomagnetic poles [poles 54–57, Table 2 in Torsvik *et al.* (1996)], which are combined in Table 3.

The pole from the Callander Complex (Symons & Chaisson 1991) is based on a well-documented palaeomagnetic study, contains three cases of reverse polarity and is supported by a positive baked contact test (pole 1 in Table 3).

The palaeomagnetic study of the Catoctin Volcanic Province (Meert *et al.* 1994) revealed two stable components of magnetization. These are Catoctin A (with steep inclination) and Catoctin B (with shallow inclination). Both components are bipolar. The fold test of McFadden (1990) is positive at the 99 per cent confidence level for Catoctin B but only at the 92 per cent confidence level for Catoctin A. Nevertheless, the authors interpreted Catoctin A as primary (pole 2 in Table 3). This conclusion is based on the fact that Catoctin B coexists in some samples with the less stable component C, whereas Catoctin A does not. Also, Catoctin B lies close to the Late Cambrian direction. However, it should be noted that all Laurentian poles in the time span 723–830 Ma (Torsvik *et al.*

Table 3. Two possible interpretations of Vendian palaeomagnetic poles from Laurentia.

N	Rock unit	Age (Ma)	North pole		Reference
			°N	°E	
(a) Selected poles following Torsvik *et al.* (1996)					
1	Callander Complex	575	−46	121	Symons & Chaisson 1991
2	Catoctin A (recalculated)	575	−43	118	Meert *et al.* 1994
3	Sept Iles Intrusion B	575	−44	135	Tanczyk *et al.* 1987
4	Long Range Dyke	615	−69	170	Murthy *et al.* 1992
(b) Alternative set of poles as discussed in the text					
5	Catoctin B (recalculated)	575	−4	183	Meert *et al.* 1994
6	Sept Iles Intrusion A	575	20	141	Tanczyk *et al.* 1987
7	Long Range Dykes	615	−11	162	Murthy *et al.* 1992

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

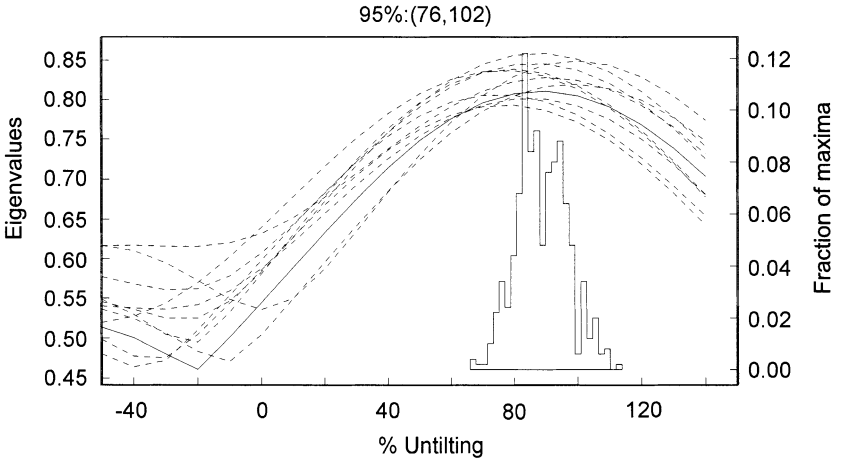


Figure 9. The result of the fold test of Tauxe & Watson (1994). See text for details.

1996) also lie close to the Late Cambrian direction. Therefore, we find the arguments put forward by Meert *et al.* (1994) to be unconvincing. Catoctin B (pole 5 in Table 3) is just as likely to be primary as Catoctin A.

The Sept Iles Intrusion B pole is a result of the reinterpretation of the original data (Tanczyk *et al.* 1987) made by Symons & Chaisson (1991). Tanczyk *et al.* (1987) suggested that Component A (pole 6 in Table 3) was primary, since it has been found only in the main intrusion and never in younger dykes, which carry Component B only. The rigorous baked contact test shows that Component A of the main intrusion gives way to Component B of the younger dykes as the dykes are approached. Symons & Chaisson (1991) made their choice in favour of the B component on the basis of its dual polarity (unlike the A component) and the fact that it differed from existing younger directions. Also, they made a 10° tilt correction to Component B (pole 3 in Table 3). However, Tanczyk *et al.* (1987) claimed that 'no post-emplacement tilting appears to be present'. Again, as in the case of the Catoctin pole, Component A is just as likely to be primary as Component B.

Murthy *et al.* (1992) carried out a palaeomagnetic study of the Long Range dykes of Labrador, Canada. The main part of the collection (dykes 2, 4 and 6; 12 sites, 69 samples) gave the north palaeomagnetic pole of 11°S, 164°E [pole 51 in Table 2 of Torsvik *et al.* (1996)]. This direction is interpreted as primary based on the positive baked contact test and was originally assigned an age of 615 ± 2 Ma—the age of dyke 1 (Kamo *et al.* 1989), whose palaeomagnetic direction was assumed to be anomalous. However, Meert *et al.* (1994) reinterpreted this result, assuming the direction of dyke 1 to be primary. Kamo & Gower (1994) have subsequently obtained an age of 615 Ma based on U–Pb baddeleyite and zircon from dyke 4, a result that is much more likely to reflect the correct age of these dykes. Therefore, it now seems more reasonable to assign an age of 615 Ma to the result from the main collection of Murthy *et al.* (1992) (pole 7 in Table 3).

The discussion above shows that the interpretation of Vendian palaeomagnetic poles for Laurentia is ambiguous. In Table 3 the primary poles as interpreted by Torsvik *et al.* (1996) are compared with an alternative interpretation. The first set of poles places Laurentia in high latitudes between 615 and 575 Ma (Table 3a), whereas it occupied low latitudes before and after this time interval. (Torsvik *et al.* 1996). The second set of poles (Table 3b), however, places Laurentia in low latitudes between 615 and 575 Ma. In this interpretation, Laurentia occupied low latitudes throughout the Vendian, and therefore the reconstructions of the palaeopositions of Laurentia and Siberia at this time as proposed by Torsvik *et al.* (1996) and Smethurst *et al.* (1998) are changed significantly.

Using the interpretation of Table 3(b), Fig. 11 shows the palaeopositions of Laurentia and Siberia in the Vendian and Early Cambrian. The mutual orientation of the two plates is in good agreement with that proposed by Hoffman (1991), Pelechaty (1996) and Dalziel (1997). It differs from the fit of Condie & Rosen (1994) by 30°–60° of rotation (unfortunately these authors provide no Euler rotation parameters), and completely disagrees with the fits of Frost *et al.* (1998) and Rainbird *et al.* (1998). Pelechaty (1996) gives some stratigraphic evidence that the rifting between Laurentia and Siberia started in the Early Cambrian, so that the connection between these plates would have existed throughout the whole of the Vendian and the earliest Cambrian. Unfortunately, there is a gap in the Siberian palaeomagnetic data between ~580 and ~550 Ma (Fig. 11). Because of the uncertainty in the age of the Lena red sediments (this study), palaeoreconstructions for the Vendian and latest Vendian–Early Cambrian have been made on the following basis: (1) the two plates are assumed to be connected throughout this time; (2) the resulting positions must fit reasonably with the palaeomagnetic data. After a few iterations the two reconstructions shown in Fig. 12 were obtained. The corresponding Euler poles of rotation are shown in Table 4.

Table 4. Euler poles of rotation with respect to the absolute framework.

Plate	Age	Latitude °N	Longitude °E	Angle (°)
Laurentia	Latest Vendian–Early Cambrian	0.9	76.0	105.9
Laurentia	Vendian	0.0	78.0	81.0
Siberia	Early Cambrian	10.0	247.0	−119.0
Siberia	Vendian	9.0	241.0	−97.0

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

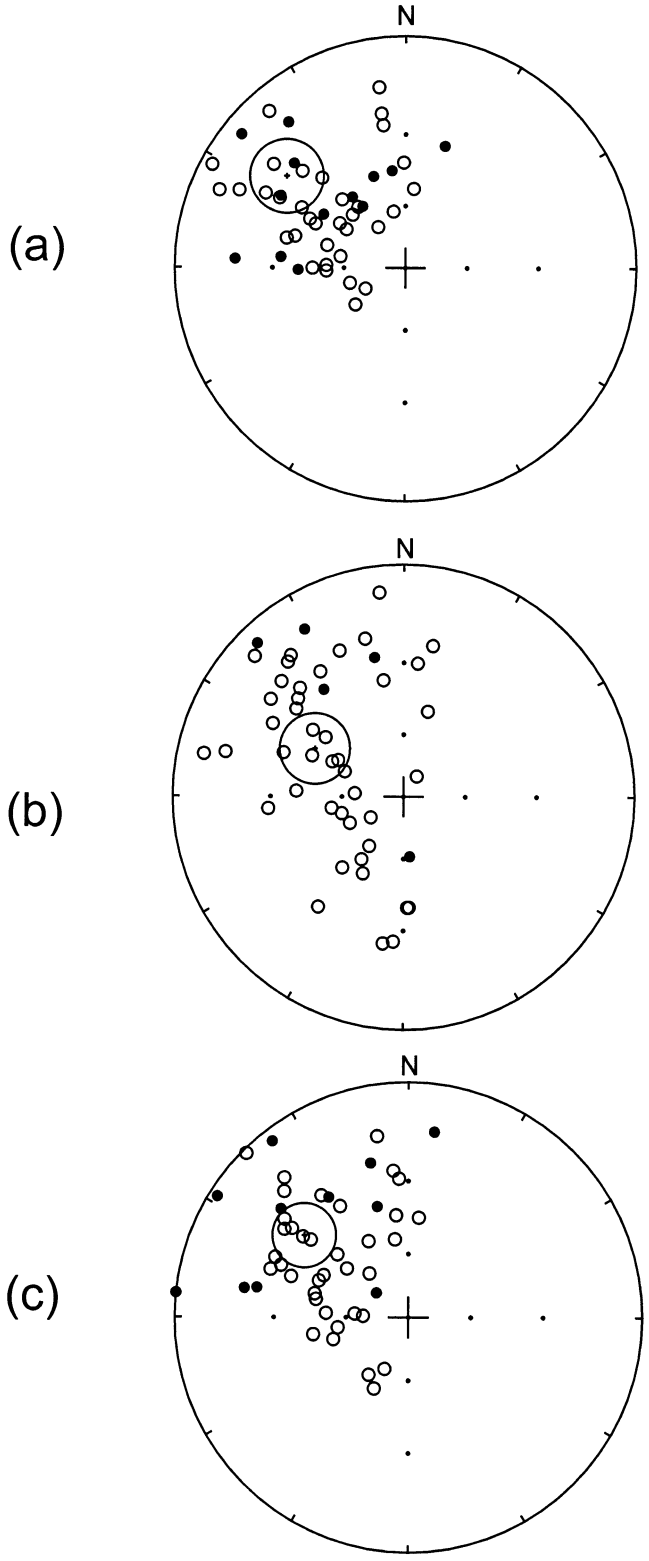


Figure 10. Stereoplots (equal-angle projection) of the 'medium-temperature component': (a) *in situ*; (b) after 100 per cent unfolding; (c) after 35 per cent unfolding. See caption to Fig. 6.

The proposed latest Vendian–Early Cambrian reconstruction is in good agreement with the 545 Ma reconstruction of Dalziel (1997), but disagrees with the 550 Ma reconstruction of Torsvik *et al.* (1996), who suggested a wide separation between Laurentia and Baltica. However, the palaeolatitudinal and azimuthal positions of these two plates are close to that shown in Fig. 11. Our Vendian reconstruction is similar to the Early to Mid-Neoproterozoic reconstruction of Dalziel (1997) and, to some extent, the 650 Ma reconstruction of Torsvik *et al.* (1996), but disagrees with the 580 Ma reconstruction of Torsvik *et al.* (1996). The latter is quite understandable because of the use of the alternative palaeomagnetic Laurentian data set of Table 3(b).

There are two possible scenarios of the plate tectonic movements of Siberia and Laurentia in the Neoproterozoic. Both indicate the low-latitude positions of these plates between $\sim$850 and $\sim$650 Ma in accordance with the available palaeomagnetic data. According to the scenario proposed by Torsvik *et al.* (1996), these plates became separated. Siberia remained at low latitudes while Laurentia moved to the southern near-polar zone (at $\sim$580 Ma) and returned to the equatorial zone at $\sim$550 Ma (Torsvik *et al.* 1996). Our scenario proposes that both plates were connected at low latitudes up to the Early Cambrian. However, the lack of palaeomagnetic data from Siberia for the interval $\sim$580–550 Ma still leaves open the possibility of a high-latitude position at that time. Further palaeomagnetic studies of this time interval in Laurentia and Siberia would be very helpful in resolving this problem.

6 CONCLUSIONS

(1) New palaeomagnetic results from red sediments in the Lena River section (Cisbaikalia) have provided evidence in favour of the low-latitude position of Siberia at some time during the Vendian (between 650 and 580 Ma).

(2) Palaeomagnetic data from Laurentia for the Vendian are open to two possible interpretations. Previously it was proposed that Laurentia occupied both low- and high-latitude positions during the Vendian. An alternative interpretation of the data set would maintain Laurentia in low latitudes throughout the Vendian.

(3) The palaeolatitudinal and azimuthal positions of Laurentia and Siberia in the Vendian and Early Cambrian based on the alternative data set from Laurentia support the palaeoreconstructions of Hoffman (1991), Pelechaty (1996) and Dalziel (1997).

ACKNOWLEDGMENTS

We are very grateful to Professor M. W. McElhinny for the informal review of our manuscript. We also thank Dr D. Evans and Dr P. Schmidt for their reviews. SAP acknowledges the University of Western Australia Gledden Senior Visiting Fellowship. This research was supported by the Australian Research Council (ARC) through its Research Centres program. Reconstructions were made with the GMAP program of T. Torsvik and M. Smethurst (NGU) and the PLATES program from the University of Texas at Austin. The IAPD program of T. Torsvik and M. Smethurst was used for calculations and several programs of P. McFadden and L. Tauxe were used for reversal and fold tests. Tectonics Special Research Centre publication no. 76.

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

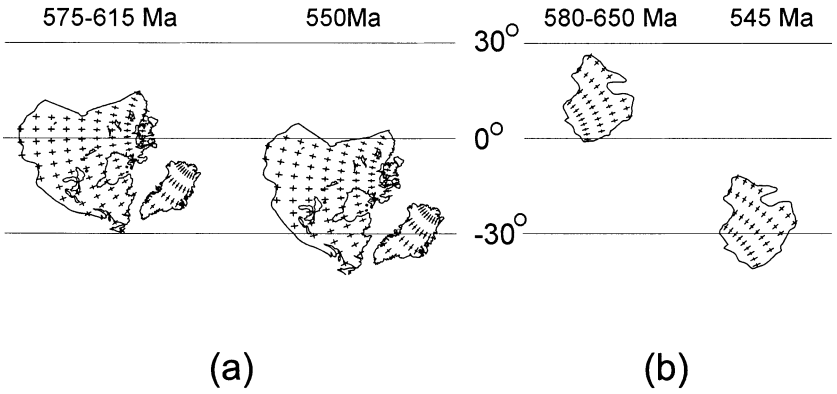


Figure 11. Palaeopositions of (a) Laurentia and (b) Siberia in the Vendian–Early Cambrian in Mercator projection. Plotted with the assistance of GMAP and PLATES.

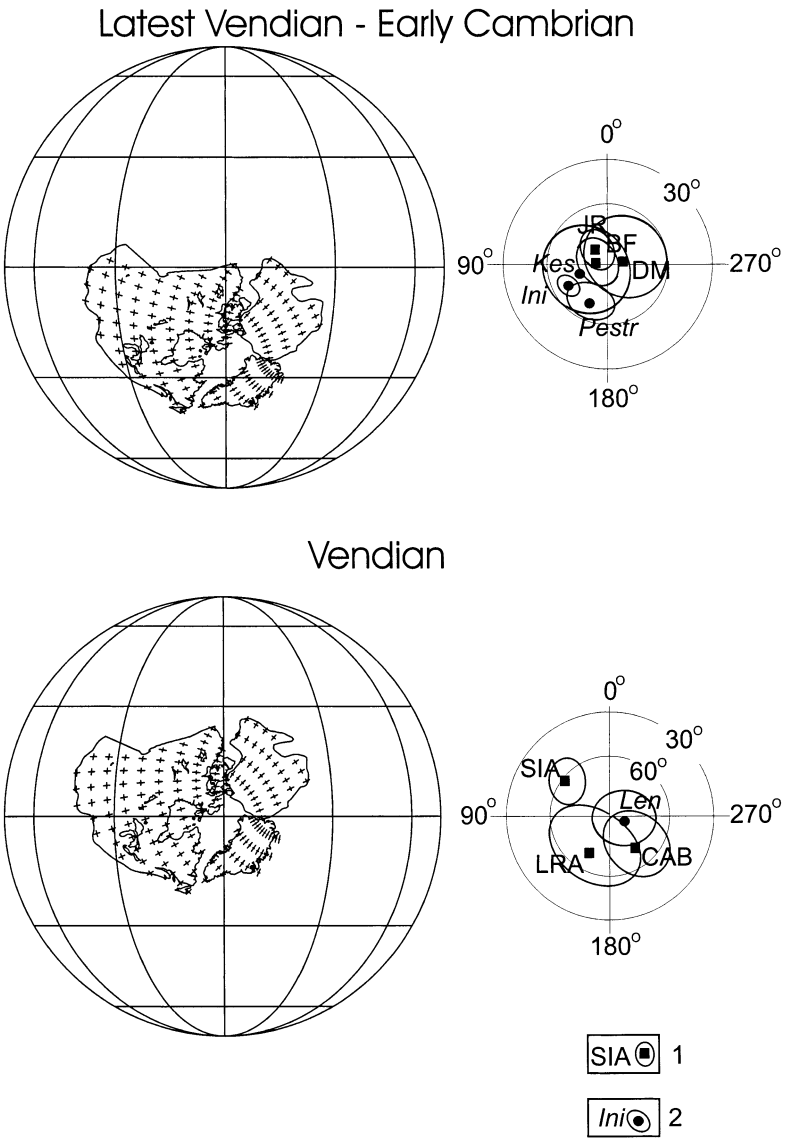


Figure 12. Proposed palaeoreconstructions of Laurentia and Siberia in Vendian and the Latest Vendian–Early Cambrian. Euler poles of rotation are summarized in Table 4. Corresponding positions of rotated north palaeopoles are shown to the right of each reconstruction. Orthogonal projection is used. 1: Laurentian palaeopoles [BF: Buckingham Flows (Dankers & LaPointe 1981); JR: Johnnie Rainstorm (Van Alstine & Gillet 1979); DM: Double Mer (Murthy *et al.* 1992); SIA: Sept Iles Intrusion A (Tanczyk *et al.* 1987); LRA: Long Range Dykes A (Murthy *et al.* 1992 with age correction of Kamo & Gower 1994); CAB: Catoctin B (Meert *et al.* 1994)]. 2: Siberian palaeopoles [Kes: Kessyusa Formation (Pisarevsky *et al.* 1997); Ini: Inican sediments (Osipova, quoted in Smethurst *et al.* 1998); Pestr: Pestrotsvetna Group (Osipova, quoted in Smethurst *et al.* 1998); Len: Lena red sediments (this study)]. Plotted with assistance of GMAP and PLATES.

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026

REFERENCES

Condie, K.C. & Rosen, O.M., 1994. Laurentia-Siberia connection revisited, *Geology,* **22,** 168–170.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antarctica-Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent, *Geology,* **19,** 598–601.

Dalziel, I.W.D., 1997. Neoproterozoic-Paleozoic geography and tectonics: review, hypothesis, environmental speculation, *Geol. Soc. Am. Bull.,* **109,** 16–42.

Dankers, P. & LaPointe, P., 1981. Palaeomagnetism of Lower Cambrian volcanics and a cross-cutting Cambro-Ordovician diabase dyke from Buckingham (Quebec), *Can. J. Earth Sci.,* **18,** 1174–1186.

Dunlop, D.J., 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis, *Geophys. J. R. astr. Soc.,* **27,** 37–55.

Dunlop, D.J. & Ozdemir, O., 1997. *Rock Magnetism. Fudamentals and Frontiers,* Cambridge University Press, Cambridge.

Fedorovsky, V.S., 1997. Dome tectonics in the Caledonian collision system of Western Cisbaikalia, *Geotectonics,* **31,** 483–497.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.,* A**217,** 295–305.

Frost, B.R., Avchenko, O.V., Chamberlain, K.R. & Frost, C.D., 1998. Evidence for extensive Proterozoic remobilization of the Aldan shield and implications for Proterozoic plate tectonic reconstructions of Siberia and Laurentia, *Precamb. Res.,* **89,** 1–23.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside-out? *Science,* **252,** 1409–1412.

Kamo, S.L. & Gower, C.F., 1994. Note: U-Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Range dykes, southeastern Labrador, *Atlantic Geol.,* **30,** 259–262.

Kamo, S.L., Gower, C.F. & Kroch, T.E., 1989. A birthdate for the Iapetus ocean? A precise U-Pb zircon and baddeleyite age for the Long Range dykes, S.E. Labrador, *Geology,* **17,** 602–605.

Khain, V.E., 1994. *Geology of Northern Eurasia (Ex-USSR),* Borntraeger, Berlin.

Khain, V.E. & Bozhko, N.A., 1988. *Historical Geotectonics. Precambrian,* Nedra, Moscow.

Khomentovsky, V.V., 1985. Vendian of the Siberian Platform, in *Vendian System,* Vol. 2, pp. 83–161, eds. Sokolov, B.S. & Fedonkin, M.A., Nauka, Moscow.

Kirschvink, J.L. & Rozanov, A.Yu., 1984. Magnetostratigraphy of Lower Cambrian strata from the Siberian Platform: a palaeomagnetic pole and preliminary polarity time-scale, *Geol. Mag.,* **121,** 189–203.

Kirschvink, J.L., Ripperdan, R.L. & Evans, D.A., 1997. Evidence for a large-scale reorganization of Early Cambrian continental masses by interial interchange True Polar Wander, *Science,* **277,** 541–545.

Lovlie, R., Torsvik, T., Jelenska, M. & Levandowski, M., 1984. Evidence for detrital remanent magnetization carried by hematite in Devonian red beds from Spitsbergen; palaeomagnetic implications, *Geophys. J. R. astr. Soc.,* **79,** 573–588.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.,* **17,** 159–162.

McFadden, P.L., 1990. A new fold test for paleomagnetic studies, *Geophys. J. Int.,* **103,** 163–169.

McFadden, P.L., 1998. The fold test as an analytical tool, *Geophys. J. Int.,* **135,** 329–338.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism, *Geophys. J. Int.,* **103,** 725–729.

McMenamin, M.A.S. & McMenamin, D.L.S., 1990. *The Emergence of Animals: the Cambrian Breakthrough,* Columbia University Press, New York.

Meert, J.G., Van der Voo, R. & Payne, T.W., 1994. Paleomagnetism of the Catoctin volcanic province: a new Vendian-Cambrian apparent polar wander path for North America, *J. geophys. Res.,* **99**(B3), 4625–4641.

Murthy, G., Gower, C., Tubrett, M. & Patzold, R., 1992. Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador, Canada, *Can. J. Earth Sci.,* **29,** 1224–1234.

Park, J.K., 1994. Palaeomagnetic constraints on the position of Laurentia from middle Neoproterozoic to Early Cambrian times, *Precamb. Res.,* **69,** 95–112.

Pavlov, V.E., 1994. Paleomagnetic poles of the Uchuro-Maya hypostratotype of Riphean and the Riphean drift of the Aldanian block of the Siberian Platform, *Doklady Acad. Sci.,* **336,** 533–537.

Pelechaty, S.M., 1996. Stratigraphic evidence for the Siberia-Laurentia connection and Early Cambrian rifting, *Geology,* **24,** 719–722.

Pisarevsky, S.A., Gurevich, E.L. & Khramov, A.N., 1997. Palaeomagnetism of Lower Cambrian sediments from the Olenek River section (northern Siberia): palaeopoles and the problem of magnetic polarity in the Early Cambrian, *Geophys. J. Int.,* **130,** 746–756.

Powell, C.McA., Li, Z.X., McElhinny, M.W., Meert, J.G. & Park, J.K., 1993. Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana, *Geology,* **21,** 889–892.

Rainbird, R.H., Stern, R.A., Khudoley, A.K., Kropachev, A.P., Heaman, L.M. & Sukhorukov, V.I., 1998. U-Pb geochronology of Riphean sandstone and gabbro from southeast Siberia and its bearing on the Laurentia-Siberia connection, *Earth. planet. Sci. Lett.,* **164,** 409–420.

Salop, L.I., 1973. *The Basic Stratigraphic Scale of the Precambrian,* Nedra, Leningrad.

Smethurst, M.A., Khramov, A.N. & Torsvik, T.H., 1998. The Neoproterozoic and Palaeozoic palaeomagnetic data for the Siberian Platform: from Rodinia to Pangea, *Earth Sci. Rev.,* **43,** 1–24.

Symons, D.T.A. & Chaisson, A.D., 1991. Paleomagnetism of the Callander Complex and the Cambrian apparent polar wander path for North America, *Can. J. Earth Sci.,* **28,** 355–363.

Tanczyk, E.I., Lapointe, P., Morris, W.A. & Schmidt, P.W., 1987. A paleomagnetic study of the layered mafic intrusion at Sept-Iles, Quebec, *Can. J. Earth Sci.,* **24,** 1431–1438.

Tarling, D.H. & Hrouda, F., 1993. *The Magnetic Anisotropy of Rocks,* Chapman & Hall, London.

Tauxe, L. & Watson, G.S., 1994. The fold test: an eigen analysis approach, *Earth. planet. Sci. Lett.,* **122,** 331–341.

Torsvik, T.H., 1986. Interactive analysis of palaeomagnetic data, in *IAPD User Guide,* pp. 1–74, University of Bergen, Bergen.

Torsvik, T.H., Lovlie, R. & Sturt, B.A., 1985. Palaeomagnetic argument for a stationary Spitsbergen relative to the British Isles (Western Europe) since late Devonian and its bearing on North Atlantic reconstruction, *Earth planet. Sci. Lett.,* **75,** 278–288.

Torsvik, T.H., Smethurst, M.A., Meert, J.G., Van der Voo, R., McKerrow, W.S., Brasier, M.D., Sturt, B.A. & Walderhaug, H.J., 1996. Continental break-up and collision in the Neoproterozoic and Palaeozoic—a tale of Baltica and Laurentia, *Earth Sci. Rev.,* **40,** 229–258.

Trunov, V.P., 1975. The correlation of Upper Proterozoic deposits of the Baikal–Patom boundary of the Siberian Platform, *Moscow Soc. Nat. Sci. Bull., Geol. Branch,* **50**(5), 28–38.

Van Alstine, D.R. & Gillet, S.L., 1979. Palaeomagnetism of Upper Precambrian sedimentary rocks from the Desert Range, Nevada, *J. geophys. Res.,* **84,** 4490–4500.

Van der Voo, R., 1988. Palaeozoic palaeogeography of North America, Gondwana, and intervening displaced terrains: comparisons of palaeomagnetism with palaeoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull.,* **100,** 311–324.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism,* pp. 254–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier, Amsterdam.

Downloaded from https://academic.oup.com/gji/article/140/3/598/670609 by Curtin University Library user on 29 April 2026